DEPI · FULL-STACK .NET WEB DEVELOPMENT · ROUND 5

Lumora Learning Platform

Project Planning & Management

TRACK	Full-Stack .NET Web Development – Round 5 (DEPI)	PARTNER	AST – CAI5_SWD5_S6
TEAM	Code Raiders	DATE	October 16, 2026
METHODOLOGY	Agile – MVP first, then iterative improvement	PROJECT WINDOW	October 2 – December 4, 2026

1. Executive Project Proposal

1.1 Overview

Lumora Learning Platform is a scalable e-learning platform inspired by industry leaders such as Udemy. Instructors can author, manage, and publish video-based courses. Students can enroll, track their progress, complete quizzes, and earn automated, verifiable certificates.

1.2 Strategic Objectives

- **Core Specification Compliance:** Deliver a robust platform that meets 100% of the EduHub MVP requirements using ASP.NET Core MVC, Entity Framework Core, and Microsoft Azure.
- **Value-Driven Innovation:** Complete the core functionality first, then use a dedicated 10-day sprint (November 21–30) for advanced features that raise the product's market value.
- **Optimal User Experience:** Provide a frictionless learning journey with a custom video player, a secure checkout flow, and responsive, interactive dashboards.

1.3 Scope of Work

- **Access Control:** Role-Based Access Control (RBAC) for Students, Instructors, and System Administrators.
- **Content Management:** Interactive course builder with video upload and streaming through Azure Blob Storage.
- **Student Engagement:** Progress tracking, quizzes, course ratings and reviews, and automated PDF certificates with verification codes (iTextSharp).
- **E-Commerce:** Enrollment with a mock checkout in the MVP; Stripe API integration (Test Mode) in the improvement phase.
- **Analytics & Reporting:** Admin dashboard built with Chart.js for performance and financial tracking.

2. Strategic Project Plan

The project follows an Agile methodology: deliver a Minimum Viable Product (MVP) quickly, then improve it through iterative cycles of innovation and refinement.

2.1 Timeline & Milestones

Phase	Timeline	Key Objectives & Milestones
Phase 1: Planning & Requirements	Oct 2 – Oct 16	Finalize planning documentation, task delegation, risk matrix, and KPIs.
Phase 2: System Analysis & Design	Oct 17 – Oct 30	Finalize the database design (ERD), UML diagrams (Use Case, Sequence, Class), and N-Tier architecture setup. **Milestone:** early delivery of all design artifacts.
Phase 3: Core Implementation (MVP)	Oct 31 – Nov 20	Develop course management, student workflows, enrollment with mock checkout, and the admin portal. Deploy a staging build to Azure. **Milestone:** stable release fulfilling all core DEPI requirements.
Phase 4: Improvement & Innovation	Nov 21 – Nov 30	Stripe Test Mode integration, competitive features, UI/UX polish, and codebase refactoring. Feature freeze on November 30.
Phase 5: Testing & Final Delivery	Dec 1 – Dec 4	End-to-End (E2E) integration testing, final Azure deployment, and presentation preparation. Bug fixes only.

2.2 Weekly Breakdown & Task Allocation

Week	Phase	Owners	Tasks & Success Criteria
Week 1 Oct 17 – 23	Phase 2	Members 1, 2, 6	**Tasks:** Set up Git workflow and N-Tier solution, draft ERD, configure ASP.NET Identity, agree UI conventions. **Success:** Project runs locally; ERD draft reviewed by the team.
Week 2 Oct 24 – 30	Phase 2	Members 1, 2, 6	**Tasks:** Finalize ERD, UML diagrams, seed data, and the shared layout. **Success:** ERD and UML approved; baseline database and authentication operate.
Week 3 Oct 31 – Nov 6	Phase 3	Members 2, 3, 5, 6 Member 4 prepares with mock data	**Tasks:** Course CRUD, sections/lessons, Home and Catalog pages, instructor dashboard, initial enrollment workflow. **Success:** Published courses appear in the catalog.
Week 4 Nov 7 – 13	Phase 3	Members 3, 4, 5	**Tasks:** Video upload (Azure Blob) and playback, progress tracking, quizzes, ratings and reviews, mock checkout. **Success:** Students can enroll, watch lessons, and take quizzes.
Week 5 Nov 14 – 20	Phase 3	Entire Team	**Tasks:** PDF certificates, admin dashboard and course approval, KPI charts, integration fixes, staging deployment, documentation. **Success:** Stable MVP on staging; students can complete courses and earn certificates.

Week	Phase	Owners	Tasks & Success Criteria
Week 6 Nov 21 – 27	Phase 4	Entire Team	**Tasks:** Stripe Test Mode integration, competitive features, UI/UX polish, refactoring. **Success:** Improvements merged with no regressions in the MVP flow.
Week 7 Nov 28 – Dec 4	Phases 4–5	Entire Team	**Tasks:** Finish improvements (until Nov 30), feature freeze, E2E testing, final Azure deployment, presentation preparation. **Success:** Deployment verified; presentation rehearsed.

3. Task Assignment & Roles

Tasks are decoupled to allow parallel work. Mock data is used early on so that no module blocks another.

Team Member	Responsibilities	Deliverables
Member 1 Architecture & Integration	Create the solution, set up N-Tier architecture and Dependency Injection, coordinate patterns, define contracts, resolve integration conflicts, build KPI query services, and own Azure deployment.	Organized, runnable solution with clear module conventions; KPI services; staging and final Azure deployments.
Member 2 Database & Authentication	Design the ERD, configure DbContext, set up ASP.NET Identity (Student / Instructor / Admin), implement authentication, and configure seed data.	Ready-to-use database and working authentication.
Member 3 Course & Instructor Management	Implement Course CRUD, manage sections and lessons, course metadata, and the Draft/Publish workflow. (Focused strictly on core course building).	Instructors can create, organize, and publish course content and sections.
Member 4 Student Learning Experience	Display course details, embed the video player, track progress, calculate completion, implement quizzes and assessment feedback, and add course ratings and reviews.	Students can watch lessons, take quizzes, and rate courses.
Member 5 Enrollment, Payment & Certificates	Implement enrollment, mock checkout, integrate Stripe Test Mode (Phase 4), handle video uploads and integration with Azure Blob Storage, and generate PDF certificates with verification codes.	Functional enrollment, secure media storage on Azure Blob, and automated certificate issuance.
Member 6 Shared UI & Admin Dashboard	Design the shared layout (Bootstrap), build Home and Catalog pages, construct the Instructor Dashboard (analytics/stats for instructors), build the Admin Dashboard (Chart.js), and manage the course approval workflow.	Consistent UI layouts, a working Instructor Dashboard, and an Admin Dashboard.

4. Risk Assessment & Mitigation Matrix

Priority is derived from likelihood and impact: High likelihood combined with High or Very High impact is rated **Critical**; all other listed combinations are rated **High**.

ID	Risk	Likelihood	Impact	Priority	Mitigation Actions	Response Plan
R1	**Project delivery delay due to tight timeframe**	High	Very High	**Critical**	Break tasks into small units early; freeze a minimal stable MVP; deliver weekly.	Reduce scope to the essential core path if behind schedule.
R2	**Disparity in experience levels among members**	High	High	**Critical**	Distribute tasks based on capability; conduct pair programming; cross-review code.	Redistribute specific parts or pair up two members temporarily on a blocked task.
R3	**Code conflicts and module integration bottlenecks**	High	Very High	**Critical**	Define clear contracts early; use isolated Git branches; mandate pull requests; merge frequently.	Halt new feature additions temporarily and fix integration issues before resuming.
R4	**Database schema instability during active development**	Medium	High	**High**	Review the ERD before implementation; enforce naming conventions; use EF Core Migrations.	Create a clean, updated migration; notify affected modules and re-test.
R5	**Authentication flaws and role permission misconfigurations**	Medium	Very High	**High**	Use server-side authorization checks; verify course ownership.	Block access to the affected route until permissions are corrected and tested.
R6	**Incomplete enrollment or unauthorized access to lessons**	Medium	Very High	**High**	Define clear registration rules; verify enrollment on the server for each request.	Disable access to paid/protected content until enrollment status is validated.
R7	**Payment gateway (Stripe) integration delays**	High	Medium	**High**	Use Stripe Test Mode; ship free enrollment and mock checkout first; integrate Stripe in Phase 4.	Deliver the MVP with free enrollment and mock checkout; keep Stripe as a deferred feature.
R8	**Exposed credentials or sensitive connection strings in Git**	Medium	Very High	**High**	Use User Secrets or environment variables; never push keys or passwords to Git.	Immediately revoke and rotate exposed keys, clean Git history, and inform the team.
R9	**Insufficient testing and bugs discovered at the final stage**	High	High	**Critical**	Write manual user scenarios; test every feature on completion; reserve the final phase (Dec 1–4) for stabilization.	Freeze all new features and fix blocking errors that disrupt the primary scenario.

5. Key Performance Indicators (KPIs)

These KPIs measure student adoption, learning engagement, course quality, financial performance, and course administration. They will be calculated with Entity Framework Core and displayed on the Admin Dashboard (ASP.NET Core MVC, Chart.js).

KPI 1 Student Enrollment Rate

Purpose	Measures student adoption: the share of registered students who have enrolled in at least one course.
Formula	(Students with at least one enrollment ÷ Total registered students) × 100
Data sources	Enrollments, Students

KPI 2 Course Completion Rate

Purpose	Measures learning engagement: the percentage of enrollments that end in course completion.
Formula	(Completed enrollments ÷ Total enrollments) × 100
Definition	A completed enrollment is one where every lesson is marked as completed.
Data sources	Enrollments, Progress, Completed Lessons

KPI 3 Average Course Rating

Purpose	Measures student satisfaction and perceived course quality from ratings submitted by students.
Formula	Sum of all ratings ÷ Number of ratings
Data sources	Reviews and Ratings

KPI 4 Monthly Revenue Growth

Purpose	Measures financial performance as the percentage change in revenue between consecutive months.
Formula	((Current month revenue − Previous month revenue) ÷ Previous month revenue) × 100
Edge case	If the previous month's revenue is zero, display "N/A" instead of a percentage.
Data sources	Successful Payments

KPI 5 Course Approval Rate

Purpose	Measures the course review process: the percentage of reviewed courses that the administrator approved.
Formula	(Approved courses ÷ Total reviewed courses) × 100
Definition	Reviewed courses are those the administrator has approved or rejected; pending courses are excluded.
Data sources	Courses, Approval Status